Light

GCSE Computer Science - Boolean Logic (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Boolean Logic

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What values can a logic gate output?

- **A.** Any number from 0 to 9
- **B.** Only 0 or 1
- **C.** True, False, or Maybe
- **D.** Positive or negative numbers

Q2 (1 mark)

An OR gate outputs 1 when:

- **A.** All inputs are 0

- B. All inputs are 1 only
- C. At least one input is 1
- D. Exactly one input is 1

Q3 (1 mark)

What happens to an AND gate's output if just one input is 0?

- A. The output is 1
- B. The output is 0
- C. The output depends on other inputs
- D. The gate stops working

Q4 (1 mark)

What does an XOR gate output when both inputs are the same?

- A. 1
- B. It depends on the values
- C. The gate produces an error
- D. 0

Q5 (1 mark)

Another name for a NOT gate is:

- A. Inverter
- B. Combiner
- C. Multiplier
- D. Splitter

Q6 (1 mark)

Each input in a truth table can have how many different values?

- A. 1 value
- B. 3 values
- C. 10 values
- D. 2 values (0 and 1)

Q7 (1 mark)

In a 2-input AND gate truth table, the output is 1 for how many rows?

- A. 0 rows
- B. 1 row
- C. 2 rows
- D. 4 rows

Q8 (1 mark)

What is the only input combination that makes an OR gate output 0?

- ○ **A.** When all inputs are 1
- ○ **B.** When inputs are different
- ○ **C.** When all inputs are 0
- ○ **D.** When exactly one input is 1

Q9 (1 mark)

The expression P.Q means:

- ○ **A.** P multiplied by Q
- ○ **B.** P divided by Q
- ○ **C.** P AND Q
- ○ **D.** P OR Q

Q10 (1 mark)

An XOR gate with inputs A = 0 and B = 1 outputs:

- ○ **A.** 1
- ○ **B.** 0
- ○ **C.** -1

- **D.** 2

Q11 (1 mark)

In Boolean algebra, what does the + symbol represent?

- **A.** Addition
- **B.** Multiplication
- **C.** OR operation
- **D.** XOR operation

Q12 (1 mark)

A NOT gate receives input 1. The output is:

- **A.** 1
- **B.** -1
- **C.** 2
- **D.** 0

Q13 (2 marks)

Both OR and XOR gates receive inputs A = 1 and B = 0. What are their respective outputs?

- **A.** OR outputs 0, XOR outputs 1

- **B.** OR outputs 1, XOR outputs 1
- **C.** OR outputs 1, XOR outputs 0
- **D.** OR outputs 0, XOR outputs 0

Q14 (2 marks)

In a circuit, inputs P and Q both pass through separate NOT gates. The two NOT outputs then go into an AND gate. If P = 1 and Q = 1, what is the final output?

- **A.** 1
- **B.** 0
- **C.** Undefined
- **D.** Depends on the AND gate

Q15 (2 marks)

Evaluate P.Q + R when P = 1, Q = 0, and R = 1.

- **A.** 1
- **B.** 0
- **C.** 2
- **D.** Undefined

Q16 (2 marks)

In a 2-input XOR truth table, how many rows show output 0?

- A. 0 rows
- B. 1 row
- C. 2 rows
- D. 4 rows

Q17 (2 marks)

What is the value of $\bar{P}$ when P = 0?

- A. 0
- B. -1
- C. P
- D. 1

Q18 (2 marks)

Inputs A and B go into an OR gate. The OR output passes through a NOT gate. If A = 0 and B = 0, what is the final output?

- A. 0
- B. 1
- C. Cannot determine
- D. Either 0 or 1

Q19 (2 marks)

Evaluate M ⊕ N when M = 1 and N = 1.

- ○ **A.** 2
- ○ **B.** 1
- ○ **C.** 0
- ○ **D.** Undefined

Q20 (2 marks)

A safe opens only when BOTH the correct code is entered AND the key is inserted. Which gate represents this logic?

- ○ **A.** OR gate
- ○ **B.** NOT gate
- ○ **C.** XOR gate
- ○ **D.** AND gate

Q21 (2 marks)

In Boolean algebra, which operation is performed first: AND or OR?

- ○ **A.** AND is performed first
- ○ **B.** OR is performed first
- ○ **C.** They are performed left to right

- **D.** They are performed simultaneously

Q22 (2 marks)

How many rows in a 2-input OR truth table have output 0?

- **A.** 0 rows
- **B.** 1 row
- **C.** 2 rows
- **D.** 3 rows

Q23 (2 marks)

Input A goes through a NOT gate, then the NOT output and input B go into an XOR gate. If A = 1 and B = 0, what is the final output?

- **A.** 0
- **B.** 1
- **C.** Cannot be determined
- **D.** Depends on XOR type

Q24 (3 marks)

A 3-input truth table has the expression A.B.C. How many rows will have output 1?

- **A.** 1 row

- B. 3 rows
- C. 4 rows
- D. 7 rows

Q25 (3 marks)

Evaluate (A + B).C when A = 0, B = 1, and C = 0.

- A. 1
- B. 2
- C. Cannot evaluate
- D. 0

Q26 (3 marks)

A fire alarm should activate when smoke is detected OR when BOTH the heat sensor triggers AND the manual button is pressed. Which expression represents this?

- A. S.H.M
- B. S + H + M
- C. S + (H.M)
- D. (S + H).M

Q27 (3 marks)

For the expression A + B + C (3-input OR), how many rows have output 1?

- **A.** 1 row
- **B.** 4 rows
- **C.** 7 rows
- **D.** 8 rows

Q28 (3 marks)

Evaluate $\bar{A}.B + C$ when A = 1, B = 1, C = 0.

- **A.** 1
- **B.** 0
- **C.** 2
- **D.** Undefined

Q29 (3 marks)

A circuit has inputs A, B, and C. A and B go into an AND gate. The AND output and C go into an XOR gate. When A = 1, B = 1, C = 1, what is the final output?

- **A.** 1
- **B.** 0
- **C.** Cannot determine

- **D.** Depends on gate order

Q30 (3 marks)

A 3-input truth table shows output Q = 1 for rows (0,0,1), (0,1,0), (1,0,0). All other rows have Q = 0. The output is 1 when exactly how many inputs are 1?

- **A.** Exactly 0 inputs are 1
- **B.** Exactly 1 input is 1
- **C.** Exactly 2 inputs are 1
- **D.** All 3 inputs are 1

Total: **55** marks • Time: **30 min**

↑ Back to top

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